

Foundation (Jalapeno)

Q1. Solve the inequality $2x + 5 > 11$.

- (A) $x > 3$
- (B) $x < 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q2. Graph the solution to $x - 2 > 4$ on a number line.

- (A) $x > 6$
- (B) $x < 6$
- (C) $x \geq 6$
- (D) $x \leq 6$
- (E) $x = 6$

Q3. Write the solution to $x > 2$ in interval notation.

- (A) $(-\infty, 2)$
- (B) $(-\infty, 2]$
- (C) $[2, \infty)$
- (D) $(2, \infty)$
- (E) $[2, \infty]$

Q4. Solve the inequality $-3x < 9$.

- (A) $x > -3$
- (B) $x < -3$
- (C) $x \geq -3$
- (D) $x \leq -3$
- (E) $x = -3$

Q5. Graph the solution to $2x + 1 < 5$ on a number line.

- (A) $x < 2$
- (B) $x > 2$
- (C) $x \geq 2$
- (D) $x \leq 2$
- (E) $x = 2$

Q6. Write the solution to $x + 1 \leq 4$ in interval notation.

- (A) $(-\infty, 3)$
- (B) $(-\infty, 3]$
- (C) $[3, \infty)$
- (D) $(3, \infty)$
- (E) $(-\infty, 3]$

Q7. Solve the inequality $4x - 2 > 2$.

- (A) $x > 1$
- (B) $x < 1$
- (C) $x \geq 1$
- (D) $x \leq 1$
- (E) $x = 1$

Q8. Graph the solution to $x - 4 \geq 0$ on a number line.

- (A) $x \geq 4$
- (B) $x \leq 4$
- (C) $x > 4$
- (D) $x < 4$
- (E) $x = 4$

Open Ended Questions (FRQ)

Q9. Solve the inequality $x^2 - 4x - 3 > 0$ and write the solution in interval notation.

Q10. Graph the solution to $2x + 5 \leq 3x - 2$ on a number line and write the solution in interval notation.

Chef's Tips

- Remind students to check their solutions.
- Allow students to use graph paper for graphing.
- Emphasize the importance of labeling the number line.